
CTLS-GAN: Inducing Class Targeted Label Skew using GANs to Steer Global Model

Abdul Nafae Imran¹ Muhammad Hamza Habib²

Abstract

Federated Learning is a distributed paradigm that is constantly under the threat from malicious actors whether these attacks seek to violate the privacy of a Federated Learning system or degrade its performance. However, a defense for such attacks cannot be set up without fully exploring a vector of attack. This paper seeks to present a novel attack vector and exploit the Non-IID environment that most systems find themselves in. Our novel attack model CTLS-GAN uses Generative Adversarial Networks (GANs) to produce high quality synthetic data for specific classes in order to artificially induce a label skew to degrade global model performance while still exhibiting the behavior of a benign client in a Non-IID environment. This method proves to be effective not only against methods that deal with Non-IID data such as MOON but also against modern robust defense mechanisms including FedCC, and SignGuard demonstrating a gap in defense mechanisms.

The codebase for the experiments conducted below can be found on [GitHub](#)

1. Introduction

Federated Learning is an ML paradigm that deals with multiple clients training the same model over a network. In a standard centralized Federated Learning setup, each client trains its local model on its local dataset distribution for a certain number of epochs, and then sends its updated parameters to the server for aggregation without risking the privacy. The server then aggregates local models from all the clients and sends back the aggregated model to each client for further training. The foundational algorithm enabling this is FedAvg (McMahan et al., 2017) that still remains the standard baseline.

Federated Learning’s aggregation method is inherently vulnerable to data poisoning attacks by the clients as it assumes the legitimacy of clients. This means that malicious

clients can degrade global model performance by using attack mechanisms like data poisoning and label flipping. Prior work (Sun et al., 2026) demonstrates GANs based data poisoning where they degrade the global model performance by introducing noise across all classes indiscriminately. However, it does not explore introducing label skew by generating synthetic data to bias global model towards a certain direction. Our work addresses this gap.

In our demonstrations, multiple malicious clients pose as valid clients. Instead of legitimate data samples, these clients use a large amount of synthetic images generated by GANs to bias the local model towards a certain class. This bias then propagates to global model through aggregation. The attack is stealthier as it requires no explicit modification to the training data and instead focuses on a class targeted approach.

In this paper, we demonstrate our attack by comparing it against several aggregation techniques meant to deal with non-IID data, and byzantine clients; including FedCC (Jeong et al., 2022), MOON (Shejwalkar & Houmansadr, 2021), FedAvg (McMahan et al., 2017), and SignGuard (Xu et al., 2022). Through these multiple experiments, we demonstrated that our attack still manages to achieve its goal against all these aggregation strategies.

2. Methodology

2.1. Problem Formulation

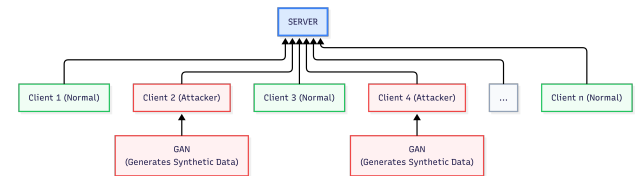


Figure 1. CTLS-GAN Setup

In traditional Federated Learning security research when it comes to degrading global model performance there is an unexplored vector of attack where label skew was utilized to achieve this goal. Most other attacks either utilize poisoning the updates or introducing back doors. In a heterogeneous

Federated Learning environment, clients naturally possess Non-IID data distributions. Our attack model CTLS-GAN makes use of GANs to exploit this feature of Federated Learning by generating high quality synthetic data for only a few specific classes. This will allow us to train the local model on the malicious clients on highly skewed data and then send their skewed gradients to the server for aggregation. As a result of this the global model will suffer with higher mis-classification rate for classes not included in the data generated by our GANs. A representation showing this setup can be seen in Figure 1.

2.2. Federated Learning Setup

We will be employing the use of one global model(server) θ_g which will use the selected aggregation method to aggregate the updates received from the local models θ_s for n clients where $n = 10$ for the purposes of this paper. From these clients m clients are adversaries they have the same local model as the benign clients θ_m where $m = 3$ for our tests. The models used by both the server and the clients consists of a simple CNN classifier Table 1.

Table 1. CifarCNN Architecture Specification

Block	Layer type	Output Shape	Kernel / Details
Input	Image	$3 \times 32 \times 32$	RGB Input
Conv Block	Conv2d	$32 \times 32 \times 32$	3×3 , pad=1
	ReLU	$32 \times 32 \times 32$	–
	MaxPool2d	$32 \times 16 \times 16$	2×2 , stride=2
	Conv2d	$64 \times 16 \times 16$	3×3 , pad=1
	ReLU	$64 \times 16 \times 16$	–
	MaxPool2d	$64 \times 8 \times 8$	2×2 , stride=2
	Conv2d	$64 \times 8 \times 8$	3×3 , pad=1
	ReLU	$64 \times 8 \times 8$	–
Classifier	Flatten	4096	–
	Linear	256	Penultimate Rep.
	ReLU	256	–
	Dropout	256	$p = 0.3$
	Linear	10	Output Logits

2.3. Dataset Setup and Configuration

For the purposes of testing the performance of our model we used the CIFAR-10 dataset with 50K images for training and 10K images for testing. We distributed the dataset among the 10 clients via Dirichlet distribution($\alpha = 0.5$) in order to simulate a natural heterogeneous environment. This allows us to first evaluate our baselines against a natural Non-IID environment before we use our attack model, this allows to make a fair comparison whether the degradation is due to the Non-IID environment or because of our induced skew.

2.4. CTLS-GAN Setup

Our attack model utilizes NVIDIA’s stylegan2-ada(Karras et al., 2020) for its GANs which has been pretrained for CIFAR-10. This GAN will produce high quality images for the targeted classes. We employ the use of two key hyperparameters β which controls the rate at which the proportion of data used for training the local models of malicious clients is increased per round and n_{top} which controls how many of the highest data classes are generated. β will allow us to increase the stealth of the attack as it prevents a high change in the updates being sent to the model. n_{top} allows us to exchange some level of detection for a higher effect on the global model. In the scenario where $n_{top} \geq 2$ the model will randomly chose which class to generate as this will allow us to ensure that not every update is the same. With it being set to 1 we will be able to significantly degrade the performance of the global model on other classes. However, it must be noted that too high of a value will result in the attack being rendered ineffective due to most classes being boosted by our GANs whereas a smaller value will result in a more concentrated attack. One advantage our attack naturally possesses is its inherent stealthiness. Since our attack simulates a benign skewed client most defenses do not have an explicit way of separating a naturally skewed client and our malicious clients in a Non-IID environment. In order to further increase stealth and prevent detection via a major difference in the training time of a local model for a malicious client we have chosen for the total data (synthetic+real) for the model the same amount as was found when the client was taken over. This ensures that a change in behavior in this regard is not detected.

For the purposes of demonstration we make a few key assumptions. Firstly we assume that a malicious actor has control of 3 out of the 10 clients (30%) and has access to the data that was stored at these clients. We also assume that the attacker has trained the GAN model on a separate machine with the data found on the clients and has made it capable of making high quality images of at least 2 classes. We will be restricting ourselves to only using the classes with the highest amount of data in one of the adversarial clients as that is what will produce the best images with GANs in the scenario where the attacker trains it.

2.5. Experiments with different Federated Learning algorithms

In order to verify the effectiveness of our attack we used aggregation/defense methods to evaluate how much performance degrades due to the attack as well as to show the attack is effective even against more modern defenses. A design showing how this setup looks like can be seen in

Figure 2.

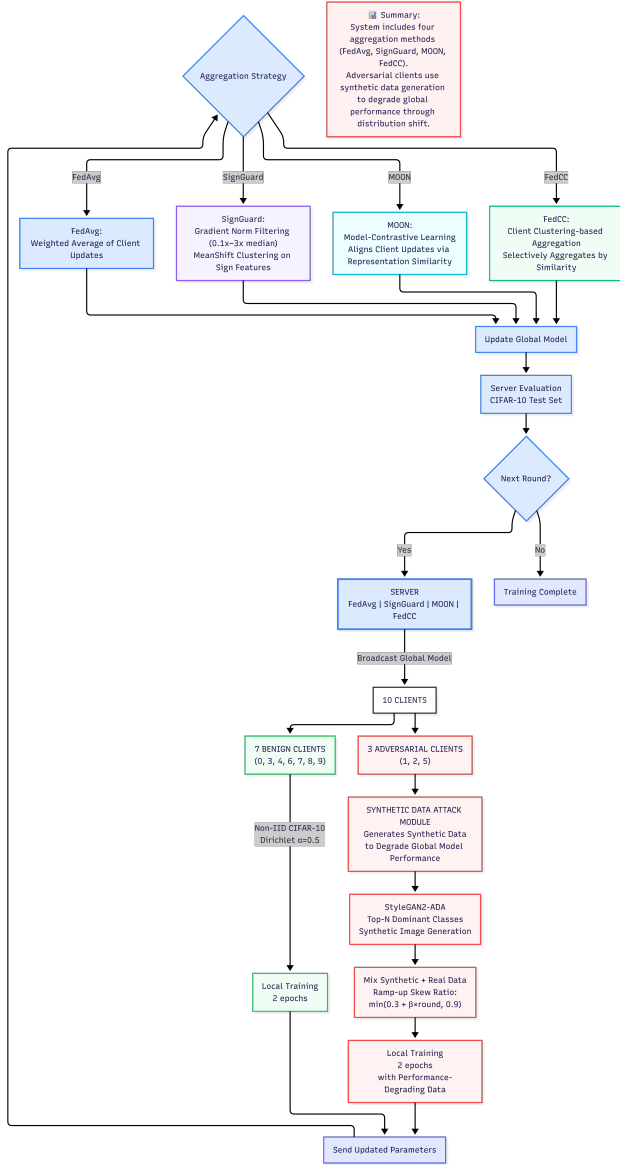


Figure 2. Representation of Experiment Setup with CTLS-GAN

2.5.1. FEDAVG

The very first aggregation method that introduced the Federated Learning field as a whole (McMahan et al., 2017) it uses simple averaging of all the weights received from the clients to update the global model. It serves as a good baseline to show how the attack acts without any special defenses or aggregation methods and allows comparison to see how other methods improve on it.

Table 2. Global Federated Learning and Attack Configurations

Parameter	Value / Selection
<i>FL Environment</i>	
Total Clients (N)	10
Compromised Clients (M)	3 (IDs: 1, 5, 2)
Communication Rounds	50
Local Training Epochs	2
Batch Size	256
<i>Optimization</i>	
Optimizer	Adam
Learning Rate (η)	0.001
Device Configuration	NVIDIA CUDA
Random Seed	67
<i>Adversarial (CTLS-GAN)</i>	
Attack Growth Rate (β)	0.12
Targeted Classes (n_{top})	1, 2

2.5.2. FEDCC

FedCC (Jeong et al., 2022) is a defense against Byzantine attacks that uses similarity (CKA score) to filter out malicious clients. It claims to deal with both Non-IID data and malicious actors. This will serve as a good baseline to see whether the attack is detectable under methods that utilize Cosine similarity.

Table 3. FedCC Configuration and Clustering Logic

Hyperparameter	Value / Selection
Extraction Layer	Penultimate ($L - 4$)
Kernel Function	RBF (Gaussian)
Kernel Bandwidth (σ)	Dynamic (Median)
Clustering Method	K-Means ($k = 2$)
Trust Heuristic	Majority Assignment
Aggregation Weight	Exp. CKA Score

2.5.3. MOON

MOON (Shejwalkar & Houmansadr, 2021) is an aggregation method designed to deal with Non-IID data and it relies on the contrast between local and global model representations to encourage local updates to stay close to the global model’s representation while pushing them away from previous local iterations. This makes it a perfect test for our attack as it is essentially inducing a heavier label skew which would make methods such as MOON a good way to ensure that our attack works even with algorithms designed to deal with Non-IID data.

Table 4. MOON Contrastive Training Hyperparameters

Hyperparameter	Value
Contrastive Weight (μ)	5.0
Temperature (τ)	0.5
Representation Dim (d)	256
Similarity Function	Cosine Similarity

2.5.4. SIGNGUARD

SignGuard (Xu et al., 2022) aims to guard against malicious clients by utilizing gradient statistics to filter out malicious clients. This is a great baseline to test our attack as it uses a different method to detect malicious actors than FedCC, and it is a modern defense that is used as a baseline for newer attack methods.

Table 5. SignGuard Defense Hyperparameters

Component	Configuration / Value
L2-Norm Bounds	$[0.1 \times \text{median}, 3.0 \times \text{median}]$
Sign-Feature Sparsity	10% of total parameters
Feature Space	Normalized Sign Ratios (Positive, Zero, Negative)
Clustering Algorithm	MeanShift (Dynamic Bandwidth)
Bandwidth Quantile	0.5
Trusted Policy	Largest Cluster Inclusion
Gradient Regularization	L2-Median Clipping

2.6. Ablation Study: Sensitivity to Attack Intensity (β)

We ran a variety of experiments to explore the effects of varying our hyperparameters. In particular, we tested FedCC and FedAvg across a variety of β values to demonstrate its effect.

 Table 6. Ablation Study: Beta (β) Sensitivity Configuration

Growth Rate (β)	Adversarial Profile / Intent
0.005	Persistence: Long-term stealth; minimal per-round gradient deviation.
0.020	Conservative: Gradual poisoning to stay within clustering thresholds.
0.050	Moderate: Balanced trade-off between detection risk and model steering.
0.120	Aggressive: (Default) Rapid injection of GAN-generated label skew.
1.000	Maximal: Immediate saturation of local training with synthetic data.

3. Results

3.1. FedAvg

When executed on FedAvg, our attack degraded its performance noticeably. The targeted attacks successfully degraded the global model’s performance by 2.5% in the one class scenario and 4.2% in two class scenarios shown in Table 7.

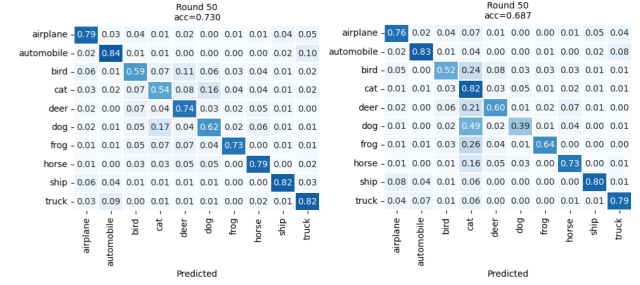


Figure 3.

Left: Confusion matrix of FedAvg classification without attack.
Right: Confusion matrix of FedAvg classification with attack.

The confusion matrix above (see figure 3) shows how the attack biases FedAvg towards a specific class, in this case, cat.

3.2. FedCC

While FedCC uses similarity (CKA score) to filter out malicious clients, our attack was stealthy enough to bypass the CKA filter most of the times. The attack reduced FedCC performance by roughly 5% in one class experiment run and 4.8% in two class experiment run as shown in table 7.

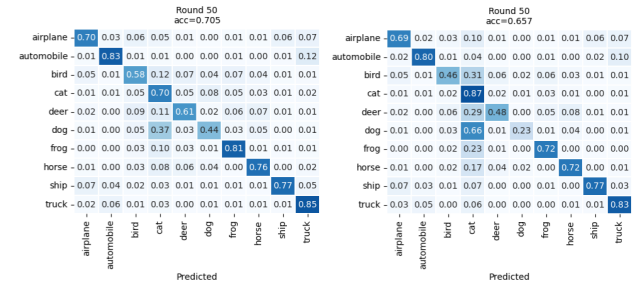


Figure 4.

Left: Confusion matrix of FedCC classification without attack.
Right: Confusion matrix of FedCC classification with attack.

The confusion matrix above (see figure 4) shows how our attack manages to bias FedCC towards the class cat.

Moreover, individual malicious clients successfully tricked

FedCC into marking them as trusted clients up to 92% of the times as shown in figure 9.

3.3. MOON

The attack was highly effective against MOON (Shejwalkar & Houmansadr, 2021). Because MOON relies on the contrast between local and global model representations, injected GAN samples severely disrupted its representation. Performance degraded by 11% during 1 class attack and 4% during 2 class attack, table 7.

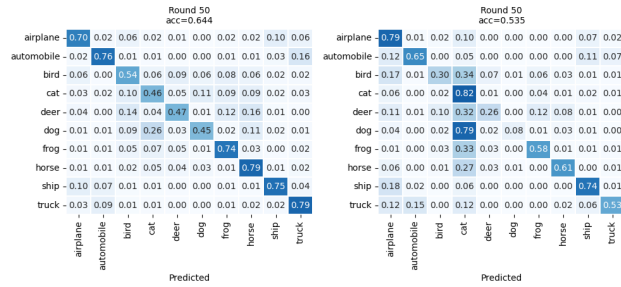


Figure 5.

Left: Confusion matrix of MOON classification without attack.
Right: Confusion matrix of MOON classification with attack.

We can see in the above (see figure 5) confusion matrix that the attack manages to bias the global model towards the cat class despite MOON trying to accommodate the skewness in the data.

STRATEGY	1-CLASS SCENARIO		2-CLASSES SCENARIO	
	NON-IID	ATTACK	NON-IID	ATTACK
FEDAVG	72.99	68.73	72.41	70.20
FEDCC	70.52	65.68	72.15	67.19
MOON	64.44	53.53	63.43	59.63
SIGNGUARD	70.38	65.08	66.25	67.28

Table 7. Comprehensive Robustness Evaluation: Comparison of Final Accuracy (%) across various strategies under 1-Class and 2-Classes adversarial settings. Results contrast baseline Non-IID performance against targeted Attack scenarios. Note: The FedAvg (Baseline) values reported are averaged across three independent runs.

3.4. SignGuard

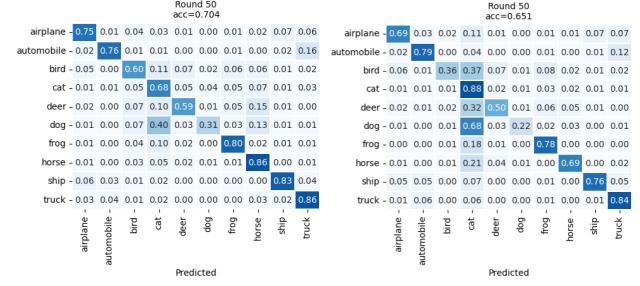


Figure 6.

Left: Confusion matrix of SignGuard classification without attack.
Right: Confusion matrix of SignGuard classification with attack.

For our second defensive strategy, we tested against SignGuard that aims to guard against malicious clients by utilizing gradient statistics. Under our attack, SignGuard suffered an accuracy loss of 5% in one class scenario and 1% in two class scenario.

3.5. Divergence & Similarity

Our attack is stealthy in a way that it manages to stay invisible in similarity matrices like L2 divergence, and Cosine Similarity.

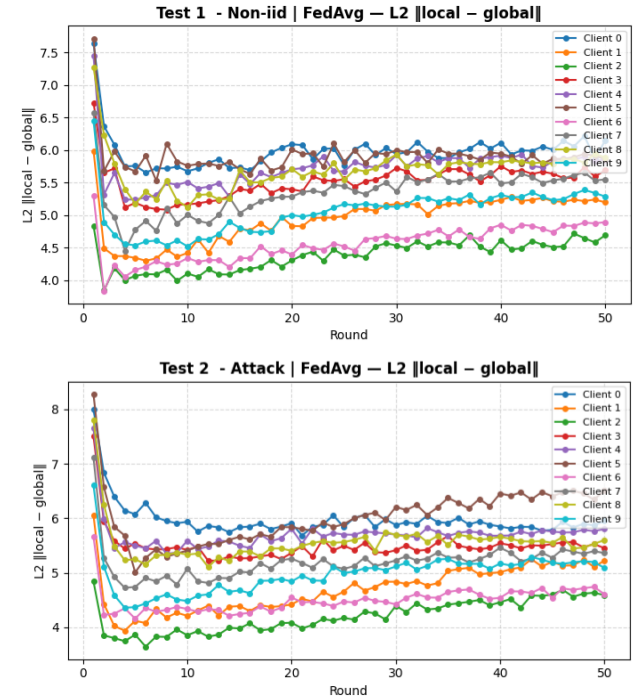


Figure 7. Left: L2 divergence of clients without attack.
Right: L2 divergence of clients during attack.

As shown in the figure 7, malicious clients [1, 2, 5] are indistinguishable from benign ones especially when compared to a normal non-IID environment which is very common in Federated Learning paradigm. As a result, when it comes to divergence, there is no way to concretely identify whether a client is malicious or benign using just divergence.

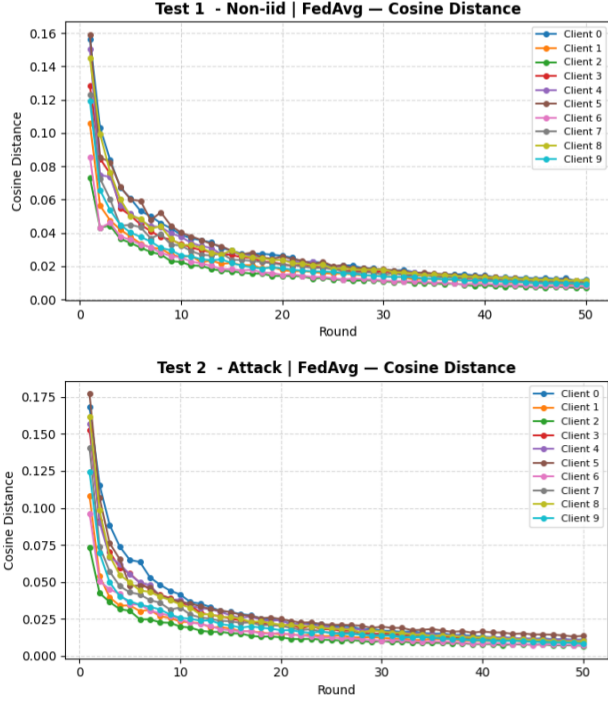


Figure 8. Left: Cosine similarity of clients without attack. Right: Cosine similarity of clients during attack.

Similarly, cosine similarity also behaves in the same way. Our malicious clients blend among the benign ones and cannot be distinguished. This stealthiness ensures that even if the attack is detected, it is difficult to tell malicious clients apart from benign ones.

3.6. Ablation Study

To find the best possible hyper-parameters for our attack, we did multiple ablation runs with different β values where β represents how quickly the attack ramps up and malicious clients replace legitimate data samples with the synthetic ones. Our results indicated that β interacts differently with different aggregations mechanisms. Since higher β value means a compromise on stealth for a aggressive attack. Since higher β values mean faster attack ramp up where legitimate data of a compromised client gets replaced by the synthetic data faster, and a lower β means a slower ramp up rate making it less prone to detection.

It must be noted that all our experiments above were con-

SCENARIO (β)	FEDAVG (%)	FEDCC (%)
BASELINE (NON-IID)	63.77	62.81
ATTACK ($\beta = 0.005$)	62.43	60.84
ATTACK ($\beta = 0.02$)	59.09	56.93
ATTACK ($\beta = 0.05$)	57.22	50.54
ATTACK ($\beta = 0.12$)	57.28	57.91
ATTACK ($\beta = 1.0$)	55.00	55.32

Table 8. Impact of Attack Intensity (β) on System Accuracy. The table compares the Final Accuracy (%) of the FedAvg and FedCC aggregation strategies across varying adversarial Dirichlet concentration parameters (β).

ducted with $\beta = 0.12$. Moreover, these ablation studies were performed on a subset of CIFAR-10, including 20k images for training and 10k for testing.

4. Discussion

4.1. Why CTLS Attack is Effective

The fundamental reason why CTLS is effective is because it mimics a naturally occurring phenomenon in Federated Learning. The label skew is expected in Federated Learning because of data heterogeneity among clients. Hence, the attack’s synthetic skew is indistinguishable from legitimate client behavior. This is evident from graphs in Figure(7 & 8) where the malicious clients blend in between normal clients.

4.2. Why it Evades FedCC

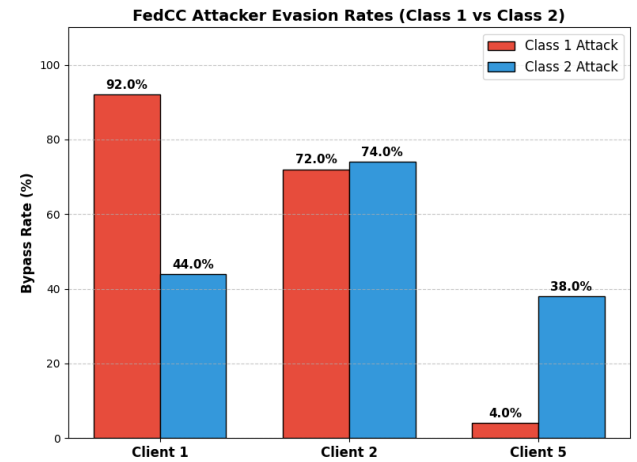


Figure 9. Bypass Rate of Malicious Clients for FedCC

FedCC relies on CKA score that clusters and filters malicious clients based on similarity. CTLS attack does not explicitly add noise or modify gradients, instead it trains on a highly skewed dataset that mimics the naturally occurring data skew, the resulting gradients are similar to those of a be-

nign client with naturally occurring skew. This renders CKA based measures unreliable. As shown in Figure 9, CTLS bypassed FedCC up to 92% of the times in case of client 1. The bypass rate varies significantly among clients suggesting that the evasion capability of a client also depends on its original data skew.

4.3. Why MOON was Most Affected

MOON suffered the most significant degradation, reducing performance by 11% (see Table 7). In MOON’s case, it pushes local representations towards global model while pushing them away from the previous local iteration. When malicious clients inject biased GAN-generated data for a target class, their representations are constantly forced towards biased direction. MOON reinforces those biases instead of correcting them because of its contrastive objective. In this way, CTLS exploits the very mechanism MOON uses to deal with skewness in data.

4.4. Why SignGuard Showed Partial Resistance

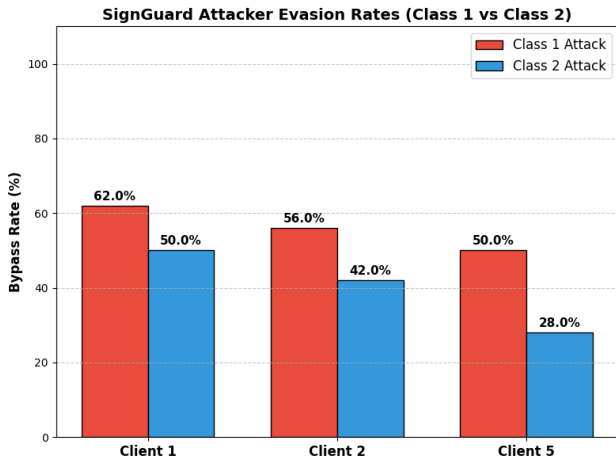


Figure 10. Bypass Rate of Malicious Clients for SignGuard

Sign guard uses gradient statistics and mean shift to detect malicious clients. The 1 class attack caused a 5% accuracy reduction but the 2 class attack couldn’t degrade accuracy. This discrepancy occurs because targeting multiple classes splits attack’s objective between two classes. As a result, the updates trigger SignGuard’s filter or they cannot form a unified adversarial cluster during mean shift aggregation. By effectively neutralizing the dispersed adversarial updates, SignGuard proved to be the most resilient among all four aggregation methods tested.

4.5. Role of β : Stealth vs. Effectiveness Tradeoff

The ablation study on β reveals a clear trade off. Very low β values preserve stealth but at the cost of effectiveness. Notably, different aggregation strategies interact differently with β . FedCC drops to 50.5% at $\beta = 0.05$ but partially rebounds at $\beta = 0.12$ (57.91%), suggesting that the clustering mechanism of FedCC somewhat adapts to aggressive injections.

4.6. Role of N-Top

The n_{top} parameter controls how many classes are targeted during the attack. For $n_{top} = 1$, the attack is focused on only one class and hence the global model is more biased towards the targeted class. The 2 class attack distributes the skew more naturally reducing per class anomaly and also diluting the target misclassifications error.

5. Conclusion

In this paper we introduced CTLS-GAN a novel method of attack in order to degrade the performance of a Federated Learning system without being able to be distinguished from a benign skewed client through exploiting the heterogeneous environment most Federated Learning systems find themselves in. Our extensive experimentation with this attack model on State-of-the-art aggregation methods (MOON) and defenses (FedCC and SignGuard) shows that current defenses do not have the capability to distinguish a benign skewed client and an attacker using our model.

Despite the success of the model future research should focus on improving its potency by either finding a more suitable β value through further ablation studies or by increasing the amount of data that is produced by the GANs model without sacrificing the stealthiness of the approach due to longer training times. Another key avenue for future work is the development of a defense mechanism for CTLS-GAN as a novel one might be required in order to confidently differentiate a benign client with skewed data and one that uses CTLS-GAN to degrade performance. Future work may also attempt to train the GANs as we assumed the attacker to do so without using a pretrained GANs as well as explore how to increase the potency of such an attack without having control of 30% of the clients or evaluating the attack against other defense mechanisms outside the ones tested in this paper.

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